How does the El Niño Southern Oscillation affect different types of cloud cover in the Maritime Continent and the Niño 3.4 region?

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1. Abstract

Previous research has identified a strong relationship between ENSO and climate in the Niño 3.4 region, particularly through interactions between sea surface temperatures and the lower atmosphere influencing cloud cover and rainfall in the equatorial Pacific. Because ENSO alters ocean temperatures and atmospheric stability in this area, it plays a major role in the formation of cloud types in this region. This investigation found that cloud types do not react uniformly to ENSO, with high cloud cover showing significantly greater modulation than low and medium cloud cover during the ENSO cycle. These differences are important as different cloud types are associated with varying extreme weather risks, such as flooding or heavy rainfall. Understanding what cloud types are likely to form during each stage of the ENSO cycle may therefore support local governments and communities to appropriately prepare for and mitigate climatic events more effectively.

2. Introduction

The tropical warm pool (TWP) is the warmest tropical ocean region located in the western Pacific and eastern Indian Oceans, within the region known as the Maritime Continent. This area influences the global climate system by producing deep convection throughout the year. This convection affects global atmospheric circulation, influencing precipitation patterns (Leung, 2022). The El Niño-Southern Oscillation (ENSO) is a naturally occurring climate phenomenon in which oceanic and atmospheric processes interact to influence global climate patterns. This cycle involves El Niño, La Niña and ENSO-neutral stages.

The El Niño stage is characterised by weakening or reversal of equatorial trade winds, which transports warm water from the TWP East towards South America and the Niño 3.4 region and changes conditions at the ocean-atmosphere boundary. The Niño 3.4 region is an area within the east-central equatorial Pacific Ocean critical to monitoring the ENSO cycle. Climate patterns are altered in the TWP and the Niño 3.4 region as changes in sea surface temperature (SST) affect the stability of the lower atmosphere.

During the El Niño phase in the Niño 3.4 region, warm, moist air rises due to buoyancy until it reaches the cloud condensation level (CCL). At this altitude, the air becomes saturated and condenses, leading to increased cloud formation and precipitation (Cai, 2020). Condensation is further enhanced by sodium chloride (NaCl) particles from sea spray in the lower atmosphere, which act as cloud condensation nuclei (CCN). In contrast, cloud cover decreases over the TWP, where lower sea surface temperatures (SSTs) stabilise the atmosphere and inhibit rising air parcels from reaching the CCL, as shown in Figure 1.

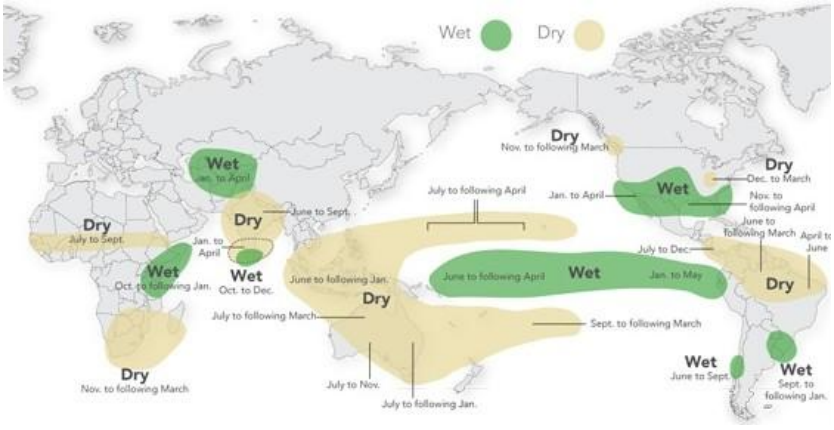


Figure 1: World map showing typical rainfall during an El Niño year (Why do we care about El Niño and La Niña?, 2025).

The increased rainfall observed over the eastern equatorial Pacific during an El Niño year is clearly shown in this figure, represented by a green band of enhanced rainfall spanning the Pacific Ocean.

The Oceanic Niño Index (ONI) is a measure of the ENSO cycle, where a persistent warming in the Niño 3.4 region of more than 0.5°C indicates an El Niño year. A cooling of more than 0.5°C is indicative of a La Niña year, whereas temperatures between the El Niño and La Niña values thresholds represent the ENSO-neutral stage of

the cycle. These values are calculated by the National Oceanic and Atmospheric Administration (NOAA). SSTs in this region are used to inform the ONI and determine the current stage in the ENSO cycle. The El Niño stage typically persists for 9–12 months (Timmermann, 2019).

This investigation tests whether ENSO phases modulate cloud fraction differently by altitude in Niño 3.4 and the Maritime Continent (2005–2025). We expect high cloud cover (HCC) to increase during El Niño in Niño 3.4, with weaker or opposite responses for medium and low cloud cover (M/LCC).

3. Methods

We used ERA5 monthly cloud fraction (TCC, LCC, MCC, HCC) for 2005–2025. Scripts extracted the NetCDF data and computed area-weighted means, monthly anomalies relative to baseline years (2010–2019), ENSO phase composites (El Niño–La Niña), and Pearson correlations with ONI/Niño 3.4. Monthly anomalies were calculated by subtracting the 2010–2019 monthly climatological mean from the corresponding monthly values over 2005–2025. These anomalies represented deviations from typical conditions for each calendar month during ENSO. ENSO composites were then derived by taking the difference between the mean El Niño anomalies and the mean La Niña anomalies, which represented the average contrast between the two ENSO phases.

ERA5 is the current-generation global atmospheric reanalysis dataset, produced by the European Centre for Medium-Range Weather Forecasts (ECMWF). NetCDF is a standard format for multi-dimensional data (Network Common Data Form). A link to the code used for this investigation can be found in the appendix.

In the ECMWF ERA5 dataset, cloud fraction represents the proportion of a grid box covered by clouds, ranging from 0 to 1. High clouds represent the upper troposphere, roughly corresponding to an altitude above six kilometres. Medium and low clouds account for the middle and lower troposphere respectively, with low clouds representing clouds below two kilometres (*ERA5 Hourly - ECMWF Climate Reanalysis*, 2026). The ONI, calculated using the SST of the Niño 3.4 region, was used to define the ENSO cycle stages. Scripts defined the Niño 3.4 region as 5°S–5°N and 170°W–120°W and the Maritime Continent region was defined as 10°S–10°N, 100°E–150°E.

4. Results

Correlation profiles for the regions (Figures 2–5) provided insight into the strength of the relationship between the ENSO cycle and cloud fraction at different altitudes in the two regions of interest. R^2 values (Table 1) showed HCC that had the strongest positive relationship with ENSO ($R^2 = 0.67$) in Niño 3.4. MCC showed a moderate positive relationship ($R^2 = 0.44$) and LCC a weakly negative relationship ($R^2 = 0.27$) in the same area. The Maritime Continent showed weaker relationships, with HCC showing a moderate negative correlation ($R^2 = 0.40$), MCC weak negative correlation ($R^2 = 0.39$) and LCC showing a negligible correlation with ENSO ($R^2 = 0.06$).

Global anomaly values quantified cloud fraction modulations in the regions during ENSO and reflected trends seen in the correlation calculations. HCC showed the largest fluctuations during the ENSO cycle in both regions, increasing in the Niño 3.4 region during El Niño and decreasing in the Maritime Continent. MCC had a more moderate response in both regions. LCC showed decreases in cloud fraction in both regions during El Niño to a lesser extent, with more pronounced responses over the Niño 3.4 region than the Maritime Continent for all cloud types. Patterns observed during El Niño were the opposite of those seen during the La Niña phase of the ENSO cycle.

Table 1: R^2 values for different types of cloud cover in the Maritime Continent and Niño 3.4 regions (2005-2025, n = 240 months)

Cloud cover type	*Maritime Continent (R^2)*	*Niño 3.4 (R^2)*
Total cloud cover (TCC)	0.40	**0.56***
High cloud cover (HCC)	0.39	**0.67***
Medium cloud cover (MCC)	0.23	**0.44***
Low cloud cover (LCC)	0.06	0.27

**Represents a positive correlation with ONI, indicating increased cloud fraction during El Niño and decreased cloud fraction during La Niña. Negative correlations indicate the opposite pattern.*

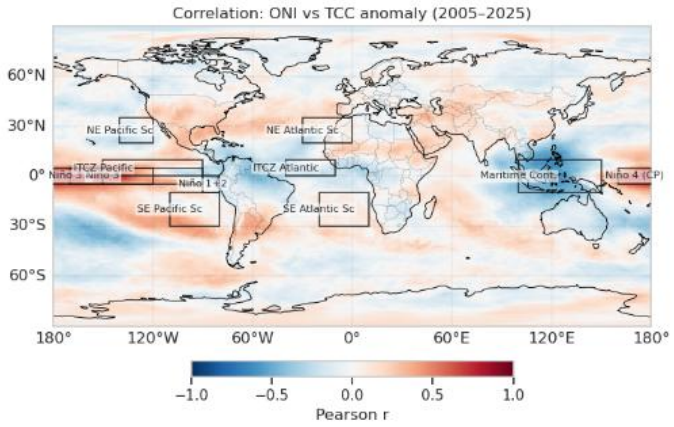


Figure 2: Monthly Pearson correlation between ONI and TCC anomalies (2005–2025, n = 240 months).

The consistency of El Niño TCC change is indicated. Red areas show positive correlations between TCC and ONI (El Niño phase), whereas blue areas represent negative correlations.

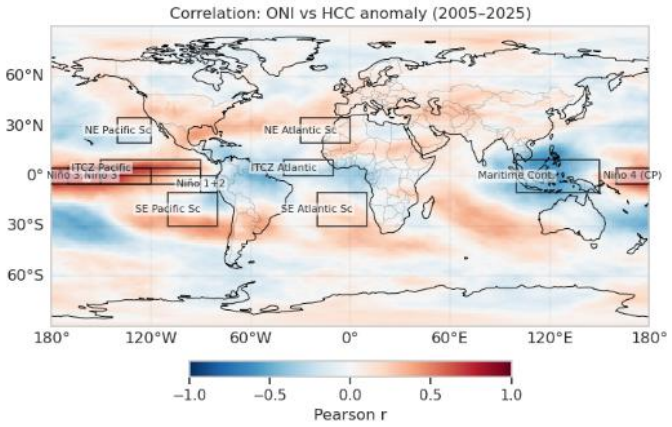


Figure 3: Monthly Pearson correlation between ONI and HCC anomalies (2005–2025, n = 240 months).

The deeper red in the Niño 3.4 region represents a stronger positive correlation with the ENSO cycle. The Maritime Continent shows a moderate negative relationship between ONI and HCC.

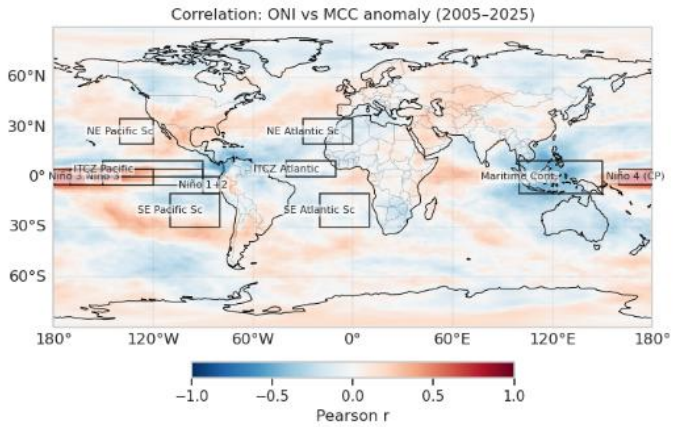


Figure 4: Monthly Pearson correlation between ONI and MCC anomalies (2005–2025, n = 240 months).

MCC has a moderate positive correlation with ONI in Niño 3.4 and a weak negative correlation in the Maritime Continent.

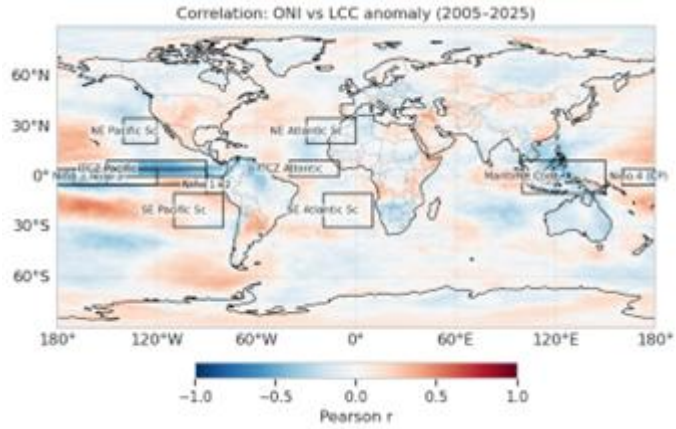


Figure 5: Monthly Pearson correlation between ONI and LCC anomalies (2005–2025, n = 240 months).

A negative correlation is observed in the Niño 3.4 region with a very weak negative relationship in the Maritime Continent.

5. Discussion

Results of this investigation showed the ENSO cycle affects different cloud types in distinct ways across the Pacific Ocean. In the Niño 3.4 region, HCC exhibited the strongest and most consistent response from 2005–2025, with Figure 3 showing substantial increases in cloud fraction during El Niño years and a relatively strong positive correlation ($R^2 = 0.67$). Figure 4 highlighted that MCC also increased during El Niño, though the relationship was weaker ($R^2 = 0.44$). Contrastingly, patterns seen in Figure 5 showed LCC decreased in El Niño years over the same region, supported by a weak negative correlation ($R^2 = 0.27$). R^2 values represent a linear correlation, meaning they can be interpreted as the proportion of change in one variable explained by the other variable. In this case, an R^2 value of 0.67 would indicate approximately two-thirds of cloud fraction variability could be attributed to the ENSO cycle, with a third of the observed fluctuations due to other factors.

Cloud-cover responses over the Maritime Continent were generally less pronounced. Some negative relationships with ENSO were observed, with R^2 values of 0.40, 0.39, 0.23, and 0.06 for TCC, HCC, MCC and LCC, respectively, indicating that ENSO's influence was weaker and more variable in this region. As in the Niño 3.4 area, HCC showed the clearest response, while correlations for MCC were weak and LCC correlations were negligible.

These differences were likely due to interactions between oceanic and atmospheric conditions. In the neutral and La Niña phases of ENSO, the easterly equatorial trade winds kept the warm water of the tropical warm pool confined to the region surrounding the Maritime Continent. During El Niño, the reversal/weakening of these winds allowed this warm water to travel across the Pacific towards the Niño 3.4 region. This reversal is illustrated in Figure 6.

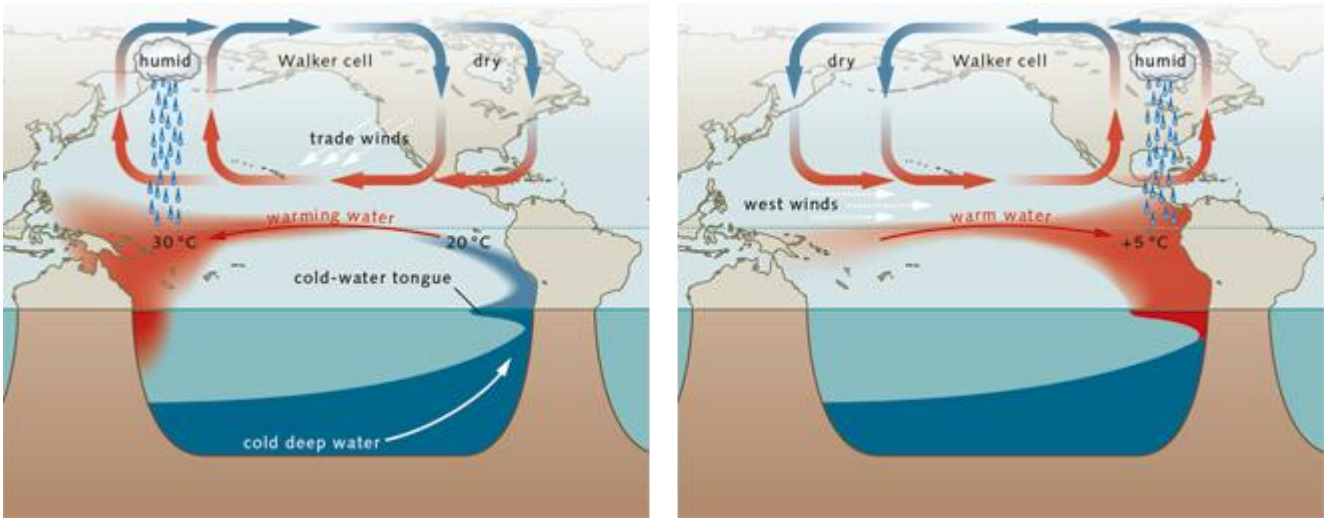


Figure 6: Diagram showing the reversal of equatorial trade winds during El Niño. This reversal allows water from the TWP to move eastwards towards Niño 3.4, increasing SSTs (How El Niño forms, 2017).

This warming increased SSTs within Niño 3.4, increasing lower atmosphere temperature and capacity to hold moisture. This created a more unstable lower atmosphere with higher moisture content. As a result, the convective available potential energy (CAPE) was elevated for air parcels in the lower atmosphere. Therefore, warm, moist air parcels experienced a greater degree of elevation. This meant the CCL was at a higher altitude as the increased initial temperature of the air parcels meant more cooling was required before saturation occurred. Water molecules then condensed onto the sodium chloride CCN from sea spray, forming clouds at a higher altitude. Consequently, HCC was favoured under El Niño conditions in Niño 3.4.

During El Niño cooler conditions caused by reduced SSTs in the Maritime Continent caused a decrease in TCC. Lessened moisture carrying capacity due to lower temperatures led to reduced cloud fraction. Some clouds that did form could have been transported away toward the Niño 3.4 region by the westerly equatorial winds associated with El Niño. These conditions particularly affected higher-altitude clouds as cooler air reached

saturation earlier. This lowered the CCL, preventing air parcels from rising high enough to be classified as HCC. This aligned with the correlation results, which showed only a minimal ENSO-related decrease in LCC (R^2 = 0.06) but a more pronounced effect on MCC and HCC (R^2 = 0.23 and 0.39, respectively).

Understanding which categories of cloud formation are likely during El Niño events is valuable for extreme weather mitigation, as different cloud types are associated with different environmental hazards.

6. Limitations

Cloud fraction across the equatorial Pacific is influenced by several factors beyond ENSO. The Maritime Continent, located along the Pacific Ring of Fire, is one of the most tectonically active regions on Earth. Volcanic eruptions release large amounts of gases and aerosols, altering atmospheric temperatures over both short and long timescales. As these processes affect cloud formation, future research could investigate how volcanic activity interacts with ENSO.

The Madden–Julian Oscillation (MJO) was not considered in this study. This intraseasonal cycle influences tropical rainfall and circulation on a 30-90 day timescale and is therefore particularly relevant for studies of short-term cloud variability (Zhang, 2005).

ERA5 cloud fraction is derived from a model-based reanalysis and may not perfectly represent observed conditions. Higher-resolution satellite data available from 1979 onwards could be used in future research to validate these results.

Finally, correlation does not imply causation. An observed relationship between two variables does not demonstrate direct causation, as both variables could be influenced by other factors. Furthermore, monthly data exhibit serial autocorrelation, meaning observations are not fully independent. This can inflate statistical significance and reduce the effective sample size, potentially overstating the strength of the results. However, the large ERA5 sample size increases statistical robustness, reducing the likelihood that results are due to chance, while still not establishing causality.

7. Conclusion

The purpose of this investigation was to discern the effects of the ENSO cycle on different altitudes of cloud cover across the Maritime Continent and the Niño 3.4 region. Results showed that cloud fraction at each altitude (high, medium and low) responded differently to El Niño events from 2005–2025. In the Maritime Continent, cloud fraction decreased during El Niño, with higher altitudes having the strongest negative correlation (R^2 = 0.39). Lower altitudes had smaller decreases in cloud cover. The Niño 3.4 region experienced greater cloud fraction during El Niño events, with higher altitudes having the strongest correlation again (R^2 = 0.67). Although total cloud cover increased in this region, low cloud cover showed a negative correlation with El Niño (R^2 = 0.27). These results supported the hypothesis that different cloud altitudes would react differently to El Niño, quantified by the correlation coefficients in Table 1. Findings from this investigation could inform local governments and communities on extreme weather mitigation, as different cloud types are associated with different atmospheric events.

Appendix

Link to Python scripts for this investigation:
https://github.com/UoM-maul1609/tutorial-schedule-1/blob/main/analyses-project/python/README.md

Acknowledgements

Generative AI was used to improve phrasing and clarity.

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